

Name:Durga Solanke Roll no-B47 Practical No 03

Aim: Data Wrangling with pandas: Manipulate and transform data to suit the analysis

1. What is data wrangling? Explain the difference between data wrangling and data preprocessing. -Data wrangling focuses on cleaning and organizing raw data, while data preprocessing focuses on preparing data for machine learning. -Data wrangling deals with missing values and structure, while preprocessing deals with scaling and encoding. -Data wrangling is used in data analysis, while preprocessing is used in model building. -Data wrangling improves data quality, while preprocessing improves model performance.
2. What is the importance of exploratory data analysis (EDA) in Pandas? EDA helps to: Understand the structure of data Detect missing values and outliers Identify patterns and relationships Check data distributions Make better decisions before modeling In Pandas, functions like .head(), .info(), .describe() help in EDA.
3. What are the main differences between Series, DataFrame, and Panel? -A Series is 1D (single column), while a DataFrame is 2D (rows and columns), and a Panel is 3D (multiple DataFrames). -Series stores single data, DataFrame stores tabular data, Panel stores multidimensional data. -Series and DataFrame are commonly used, while Panel is deprecated (not used now).
4. Explain the use of: a. isnull() b. notnull() c. dropna() d. fillna() a. isnull() Checks for missing values Returns True where data is missing df.isnull() b. notnull() Opposite of isnull() Returns True where data is present df.notnull() c. dropna() Removes rows or columns with missing values df.dropna() # removes rows with missing values d. fillna() Fills missing values with a specified value df.fillna(0)
5. What is column transformation and why is it important in data analysis? -Column Transformation Column transformation means modifying or converting data in columns to make it suitable for analysis. Examples Changing data types (int, float) Creating new columns Scaling values Encoding categorical data Importance Improves data quality Makes data consistent Helps in better analysis and modeling Converts raw data into meaningful insights

```
# Import library
import pandas as pd
# Create sample dataset
data = {
    "Name": ["Amit", "Sara", "John", None],
    "Age": [25, None, 30, 22],
    "City": ["Mumbai", "Delhi", None, "Pune"],
    "Salary": [50000, 60000, None, 45000]
}
df = pd.DataFrame(data)
print("Original Data:")
print(df)
# 1. Handling Missing Values
df["Age"].fillna(df["Age"].mean(), inplace=True)
df["Salary"].fillna(0, inplace=True)
df.dropna(subset=["Name"], inplace=True)
# 2. Renaming Columns
df.rename(columns={"Salary": "Monthly_Salary"}, inplace=True)
# 3. Changing Data Types
df["Age"] = df["Age"].astype(int)
# 4. Filtering Data
filtered_df = df[df["Monthly_Salary"] > 45000]
# 5. Adding New Column
```

```
df["Bonus"] = df["Monthly_Salary"] * 0.10
# 6. Grouping Data
grouped = df.groupby("City")["Monthly_Salary"].mean()
# Display Results
print("\nCleaned Data:")
print(df)
print("\nFiltered Data (Salary > 45000):")
print(filtered_df)
print("\nAverage Salary by City:")
print(grouped)
```

Original Data:

	Name	Age	City	Salary
0	Amit	25.0	Mumbai	50000.0
1	Sara	NaN	Delhi	60000.0
2	John	30.0	None	NaN
3	None	22.0	Pune	45000.0

Cleaned Data:

	Name	Age	City	Monthly_Salary	Bonus
0	Amit	25	Mumbai	50000.0	5000.0
1	Sara	25	Delhi	60000.0	6000.0
2	John	30	None	0.0	0.0

Filtered Data (Salary > 45000):

	Name	Age	City	Monthly_Salary
0	Amit	25	Mumbai	50000.0
1	Sara	25	Delhi	60000.0

Average Salary by City:

City	
Delhi	60000.0
Mumbai	50000.0

Name: Monthly_Salary, dtype: float64

/tmp/ipykernel_1655/2146821277.py:14: FutureWarning: A value is trying to be set on a copy of a DataFrame. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate result will be discarded.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value},

```
df["Age"].fillna(df["Age"].mean(), inplace=True)
```

/tmp/ipykernel_1655/2146821277.py:15: FutureWarning: A value is trying to be set on a copy of a DataFrame. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate result will be discarded.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value},

```
df["Salary"].fillna(0, inplace=True)
```

Conclusion

Data wrangling using Pandas helps transform raw, messy data into a clean and structured format. It improves data quality, makes analysis easier, and ensures more accurate results. Techniques like handling missing values, filtering, grouping, and feature creation are essential steps in any data analysis workflow.

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